

Linear Search

Definition

Linear Search is a simple searching algorithm that checks each element of a list **one by one from the beginning to the end** until the required element is found or the list ends.

Algorithm

1. Start from the first element of the array.
2. Compare the current element with the target element.
3. If they are equal, return the position/index.
4. If they are not equal, move to the next element.
5. Repeat until the element is found or the array ends.
6. If the element is not found, return -1.

```
Element found at index 2
...Program finished with exit code 0
Press ENTER to exit console.█
```

Output:

Time Complexity

- **Best Case:** $O(1)$ — element is found at the first position.
- **Average Case:** $O(n)$
- **Worst Case:** $O(n)$ — element is at the last position or not present.

Space Complexity

- **$O(1)$** — no extra space is required apart from a few variables.

In short:

Time: $O(n)$ worst case

Space: $O(1)$